

ASSIGNMENT

Apache Pig for Product Review Data Processing and Average Rating Analysis

Name: Sinchana.M

USN: 4GW23CI048

MLOps Assignment

2. Objective

To use Apache Pig and Pig Latin to process an e-commerce product review dataset stored in Hadoop HDFS.

The objectives are to:

1. Load review data from HDFS.
2. Remove invalid reviews having null ratings.
3. Remove reviews having ratings below 1.0.
4. Group valid reviews according to product category.
5. Calculate the average rating for each product category.
6. Store the processed results back into HDFS.
7. Display and verify the final output.

3. Problem Statement

An e-commerce company has a dataset containing product reviews and ratings. Some reviews contain invalid ratings. A rating is considered invalid when:

- The rating is null.
- The rating is below 1.0.

The task is to use Apache Pig to remove these invalid reviews and calculate the average rating for each product category.

The final result should provide the average rating of every category containing valid reviews.

4. Technologies Used

Apache Hadoop

Distributed data processing framework used for processing data.

HDFS

Used to store the review dataset and the processed output.

YARN

Used for resource and job management in the Hadoop environment.

Apache Pig 0.17.0

Used for data processing through Pig Latin.

Pig Latin

Data-flow scripting language used to load, filter, group and analyze the review data.

Linux/Ubuntu

Operating environment used to execute Hadoop and Apache Pig commands.

5. Hadoop Environment

Before executing the Pig program, the Hadoop services were started.

The following command was used to verify the running services:

Jps

The following Hadoop processes were running:

- NameNode
- DataNode
- SecondaryNameNode
- ResourceManager
- NodeManager

This confirms that both HDFS and YARN were running successfully.

6. Apache Pig Verification

Apache Pig version 0.17.0 was installed and verified using:

Pig -version

The installed version was:

Apache Pig version 0.17.0

Pig was also started successfully using:

Pig

Pig selected MapReduce as the execution engine and successfully connected to:

Hdfs://localhost:9000

This confirms that Apache Pig was correctly integrated with Hadoop.

7. Input Dataset

The input file used for the experiment is:

Reviews.csv

The dataset contains the following fields:

- Review_id – Unique ID of the review
- Product_id – Unique ID of the product
- Category – Product category
- Rating – Rating given by the customer

Dataset:

Review_id,product_id,category,rating

1,P101,Electronics,4.5

2,P102,Books,3.0

3,P103,Electronics,5.0

4,P104,Books,0.0

5,P105,Electronics,

6,P106,Books,4.0

7,P107,Clothing,2.5

8,P108,Clothing,0.0

9,P109,Electronics,3.5

Invalid Records:

4,P104,Books,0.0

5,P105,Electronics,

8,P108,Clothing,0.0

These records are removed because a rating below 1.0 is invalid and an empty rating is a null value.

8. Storing the Dataset in HDFS

An HDFS directory was created using:

```
hdfs dfs -mkdir -p /user/aiml/reviews
```

The input file was uploaded to HDFS using:

```
hdfs dfs -put -f reviews.csv /user/aiml/reviews/
```

The contents of the HDFS directory were verified using:

```
hdfs dfs -ls /user/aiml/reviews
```

The input file was then displayed using:

```
hdfs dfs -cat /user/aiml/reviews/reviews.csv
```

The HDFS input location is:

/user/aiml/reviews/reviews.csv

9. Pig Latin Program

The Pig Latin program is stored in:

Review_analysis.pig

The complete program is:

```
Reviews = LOAD '/user/aiml/reviews/reviews.csv'
USING PigStorage(',')
AS (
  Review_id:int,
  Product_id:chararray,
```

```

    Category:chararray,
    Rating:float
);

Data = FILTER Reviews BY Review_id IS NOT NULL;
Valid_reviews = FILTER Data BY Rating IS NOT NULL AND Rating >= 1.0;
Grouped_reviews = GROUP Valid_reviews BY Category;

Average_ratings = FOREACH Grouped_reviews GENERATE
    GROUP AS Category,
    AVG(Valid_reviews.Rating) AS Average_rating;

STORE Average_ratings INTO '/user/aiml/review_output'
USING PigStorage(',');

```

10. Explanation of the Pig Program

10.1 Loading the Dataset

The following statement loads the dataset from HDFS:

```

Reviews = LOAD '/user/aiml/reviews/reviews.csv'
USING PigStorage(',')
AS (
    Review_id:int,
    Product_id:chararray,
    Category:chararray,
    Rating:float
);

```

LOAD reads the data from HDFS. PigStorage(',') specifies that the fields are separated by commas. The schema defines the data types of the four fields.

10.2 Removing the Header

The following statement is used to remove the header row:

```
Data = FILTER Reviews BY Review_id IS NOT NULL;
```

The header does not contain an integer review_id, so it is removed by this filter.

10.3 Removing Invalid Ratings

The main filtering operation is:

```

Valid_reviews = FILTER Data
BY Rating IS NOT NULL AND Rating >= 1.0;

```

This performs two checks:

- Rating IS NOT NULL removes reviews without a rating.
- Rating >= 1.0 removes ratings below 1.0.

Therefore, only valid ratings are retained.

10.4 Grouping by Category

The valid records are grouped using:

```
Grouped_reviews = GROUP Valid_reviews BY Category;
```

This creates one group for each product category.

For this dataset, the categories are:

- Books
- Clothing
- Electronics

10.5 Calculating Average Rating

The average rating is calculated using:

```
Average_ratings = FOREACH Grouped_reviews GENERATE
  GROUP AS Category,
  AVG(Valid_reviews.Rating) AS Average_rating;
```

AVG() calculates the average rating for each category.

10.6 Storing the Result

The final results are stored in HDFS using:

```
STORE Average_ratings INTO '/user/aiml/review_output'
USING PigStorage(',');
```

The output directory is:

/user/aiml/review_output

11. Executing the Pig Program

The Pig program was executed using:

Pig review_analysis.pig

Pig selected MapReduce as the execution engine.

The program successfully completed with:

MapReduceLauncher – Success!

This confirms that the Pig data-processing job was successfully executed.

12. Output Verification

The generated output directory was checked using:

```
hdfs dfs -ls /user/aiml/review_output
```

The output contained:

- _SUCCESS
- part-r-00000

The final output was displayed using:

```
hdfs dfs -cat /user/aiml/review_output/part-r-00000
```

The output was:

Books,3.5

Clothing,2.5

Electronics,4.333333333333333

13. Result Analysis

Books

Valid ratings:

3.0

4.0

Average:

$$(3.0 + 4.0) / 2 = 3.5$$

Therefore:

Books = 3.5

Clothing

Valid rating:

2.5

The rating 0.0 was removed because it is below 1.0.

Average:

2.5

Therefore:

Clothing = 2.5

Electronics

Valid ratings:

4.5

5.0

3.5

The null rating was removed.

Average:

$$(4.5 + 5.0 + 3.5) / 3$$

$$= 13.0 / 3$$

$$= 4.333333333333333$$

Therefore:

Electronics = 4.333333333333333

14. Commands Used

1. Check Hadoop Services

jps

Expected services:

- NameNode
- DataNode
- SecondaryNameNode
- ResourceManager
- NodeManager

2. Go to the Assignment Directory

```
cd ~/pig-assignment
```

3. Create the Input Dataset

```
nano reviews.csv
```

Enter:

```
review_id,product_id,category,rating
1,P101,Electronics,4.5
2,P102,Books,3.0
3,P103,Electronics,5.0
4,P104,Books,0.0
5,P105,Electronics,
6,P106,Books,4.0
7,P107,Clothing,2.5
8,P108,Clothing,0.0
9,P109,Electronics,3.5
```

Save using:

Ctrl+O

Enter

Ctrl+X

Verify:

```
cat reviews.csv
```

4. Create HDFS Input Directory

```
hdfs dfs -mkdir -p /user/aiml/reviews
hdfs dfs -ls /user/aiml
```

5. Upload Dataset to HDFS

```
hdfs dfs -put -f reviews.csv /user/aiml/reviews/
hdfs dfs -ls /user/aiml/reviews
hdfs dfs -cat /user/aiml/reviews/reviews.csv
```

6. Create the Pig Script

```
nano review_analysis.pig
```

Enter the Pig Latin program shown in Section 9.

Save:

Ctrl+O

Enter

Ctrl+X

Verify:

```
cat review_analysis.pig
```

7. Remove Previous Output

```
hdfs dfs -rm -r /user/aiml/review_output
```

This is necessary because Pig does not overwrite an existing output directory.

8. Execute the Pig Program

```
pig review_analysis.pig
```

Successful execution should show:
MapReduceLauncher - Success!

9. Check the Output Directory

```
hdfs dfs -ls /user/aiml/review_output
```

Expected:
_SUCCESS
part-r-00000

10. Display Final Output

```
hdfs dfs -cat /user/aiml/review_output/part-r-00000
```

Expected output:

Books,3.5

Clothing,2.5

Electronics,4.333333333333333